Sahaj Bhadja

(213) 284-4651 sahaj.bhadja@gmail.com sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years building **TypeScript/C++ applications**, **interactive graphics**, spatial workflows, and performance-sensitive browser interfaces.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Web Technologies, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Optimized indexed dashboard queries across Azure SQL and PostgreSQL, reducing latency from 1.2 seconds to 350 milliseconds through **performance profiling, spatially structured data access, and careful correctness validation**.
- Rebuilt asynchronous FastAPI workflows with explicit job states, failure isolation, and concurrent scheduling, processing 1K+ jobs at 3x throughput while strengthening **performance, reliability, and debuggability** under sustained workloads.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built React interfaces for uploads, image review, grading history, and model controls, supporting 10K+ daily requests through **responsive browser interactions, REST APIs, and production debugging** across complex visual workflows.
- Developed concurrent Python pipelines for OCR and multimodal image processing, handling 1K+ files per batch while reducing turnaround 60% through **parallel execution, validation, and performance-oriented failure handling**.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Integrated HD-map geometry, lane direction, and ego trajectories through **spatial-temporal transforms and projections**, improving closed-loop planning stability 15% across reproducible autonomous-driving simulation baselines and controlled research evaluations.
- Built a PyTorch Decision Transformer processing 100K+ trajectory logs, improving planning 18% while analyzing **large-coordinate spatial data, numerical behavior, and visualization-ready outputs** through reproducible simulation research pipelines.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Optimized distributed APIs and data pipelines, improving response times 30% through **profiling, object-oriented design, caching, indexing, and payload reduction** while supporting reliable interaction across Windows, Linux, and macOS.
- Automated cross-platform builds and 20+ unit and integration suites using Jenkins, PyTest, Mockito, and virtual machines, strengthening **regression testing, correctness, and performance tracking** across frequent software releases.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built responsive React and TypeScript interfaces with real-time WebSocket updates, synchronizing 2M+ records across locations while improving **browser performance, interaction reliability, and maintainable component architecture** for operational users.
- Developed eight modular services and automated Docker/Jenkins releases, reducing deployment time 50% while applying **Sentry monitoring, structured debugging, automated testing, and practical shipping judgment** across production workflows.

Skills

Languages & Web: TypeScript, JavaScript, C++, Python, HTML5, CSS3, React, WebGL2, GLSL, Three.js
3D Graphics: Rendering, GPU Picking, Point Clouds, Transforms, Projections, Planes, Camera Controls, Clipping, Snapping
Spatial Systems: LiDAR, Geospatial Data, Spatial Algorithms, Geometry, Numerical Methods, CAD, Octrees
Performance: Profiling, Benchmarking, LOD, Frustum Culling, Memory Optimization, Caching, Algorithms
Engineering: Git, Docker, CI/CD, Unit Testing, Integration Testing, Debugging, WebSockets

Projects

Point Cloud Engine | *TypeScript, Three.js, WebGL2, GLSL, Octrees, Web Workers*

- Designed a **browser point-cloud renderer** using octree level-of-detail selection, frustum culling, GPU picking, shader-based coloring, and worker-driven streaming for responsive inspection of large, noisy real-world geospatial datasets.

GeoSketch | *TypeScript, WebGL2, GLSL, Computational Geometry, CAD Tools*

- Built an **interactive 3D geometry toolkit** supporting transforms, projections, clipping planes, ray intersections, snapping, construction planes, camera controls, and precision measurements with deterministic numerical tests for noisy spatial inputs.

CPU-scheduling Simulator | *C++, React, Algorithms, Performance Analysis* | Live

- Implemented **performance-sensitive C++ algorithms** for FCFS, SJF, SRT, Round Robin, priority, and multilevel queues, validating correctness and visualizing runtime behavior across 1K+ simulated process combinations with interactive controls.